

ENGINEERING DESIGN & INNOVATION (EDI)

Rescue Robotics

0th Review Presentation: Problem Formulation, System Architecture & Milestone Progress

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REVIEW STRUCTURE

Presentation Agenda

1. Introduction | 2. Problem Statement | 3. Proposed Solution | 4. Progress Level |
5. Conclusion

1. Introduction to Rescue Robotics

- **Emergency Assistance:** Search and Rescue (SAR) UGVs reach dangerous collapse zones, seismic ruins, and borewells where human entry is impossible.
- **Life-Saving Telemetry:** Integrates multi-modal sensors to detect human life signs, locate victims, and monitor toxic hazards.
- **Safety Scout:** Operates as a remote sensory extension for first responders, mitigating risks of structural re-collapse.
- **Project Objective:** Building an agile, cost-effective robotic platform tailored for rapid deployment in urban emergencies.



2. Problem Statement & Critical Gaps

Risks to Human Rescuers

Traditional search operations expose emergency workers to falling debris, gas explosions, toxic fumes, and oxygen-depleted confined voids during critical golden hours.

Lack of Visual & Environmental Data

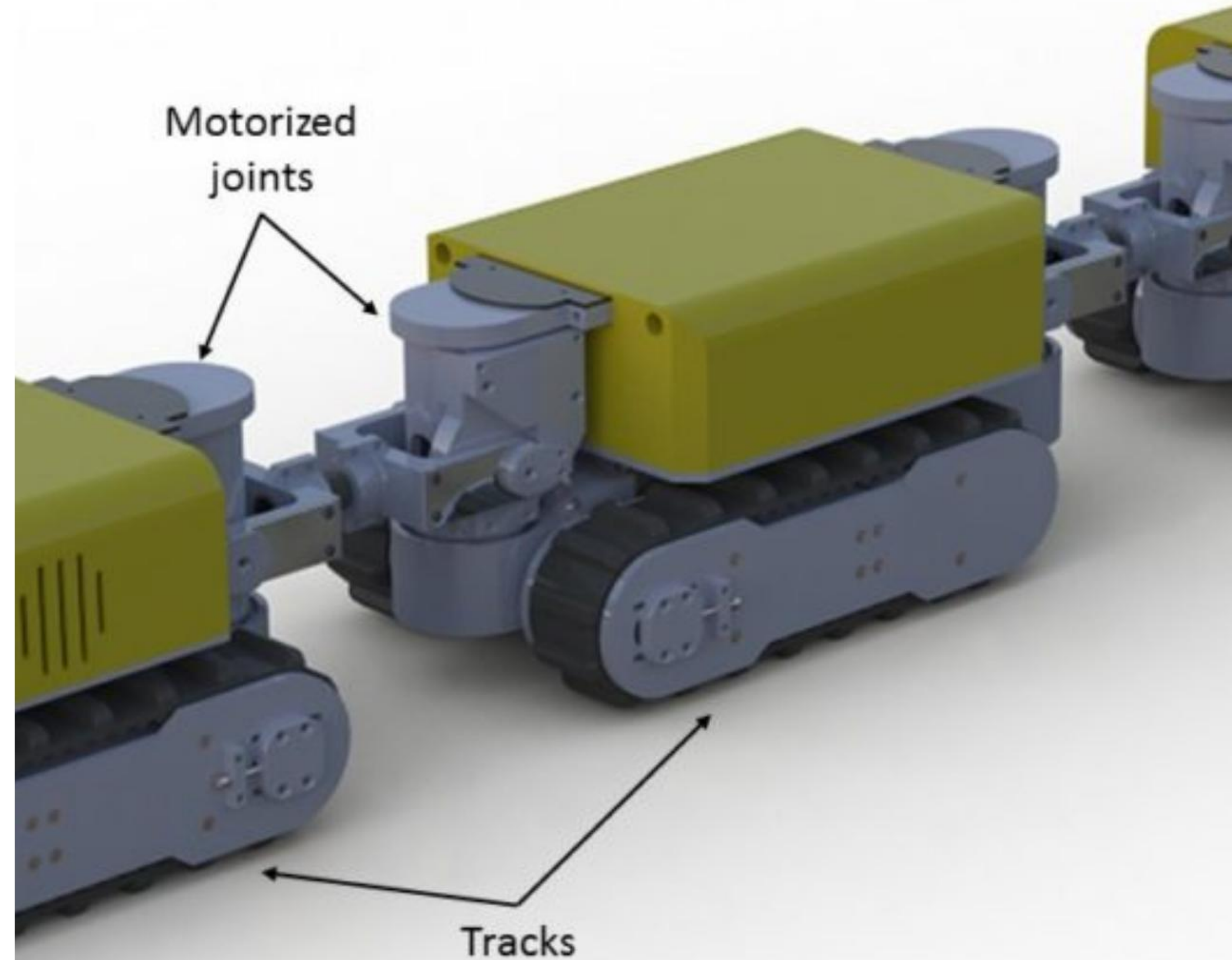
First responders lack immediate situational awareness deep inside rubble voids, leading to delayed search times and reduced survival chances for trapped victims.

3. Proposed Robotic Solution

Track-Driven UGV Platform

Our project develops a semi-autonomous Unmanned Ground Vehicle (UGV) capable of traversing highly unstable disaster terrain, steep rubble, and narrow passages.

Equipped with real-time camera streaming, thermal sensing, life-detection modules, and long-range wireless telemetry for seamless command control.



Key Technological Features



Vision & Thermal Stream

Combines optical HD video with thermal camera detection to identify body heat signatures in smoke and low-light environments.



Gas & Life Sensing

Monitors CO2 levels and hazardous gases (MQ series) to detect respiration signs while flagging toxic atmospheric risks.



Wireless Telemetry

Long-range RF and Wi-Fi mesh communication ensures steady live video and sensor data transmission through concrete barriers.

Hardware & Software Specifications

Subsystem	Components Selected	Functional Role
Processing Core	ESP32-CAM & Arduino Mega	Handles low-latency video streaming & motor telemetry controls
Actuation & Power	High-Torque DC Motors + L298N Driver	Drives track wheels across high-friction, uneven debris surfaces
Sensors Payload	MQ-135 Gas, Ultrasonic, DHT11 Temp	Provides environmental sensing, obstacle detection & life indication
Control Interface	Web-based GUI Dashboard & RF Remote	Enables real-time remote teleoperation and live visual telemetry

4. Progress Level: 0th Review Milestone

35%

Overall Project Completion

0th Review Progress Summary

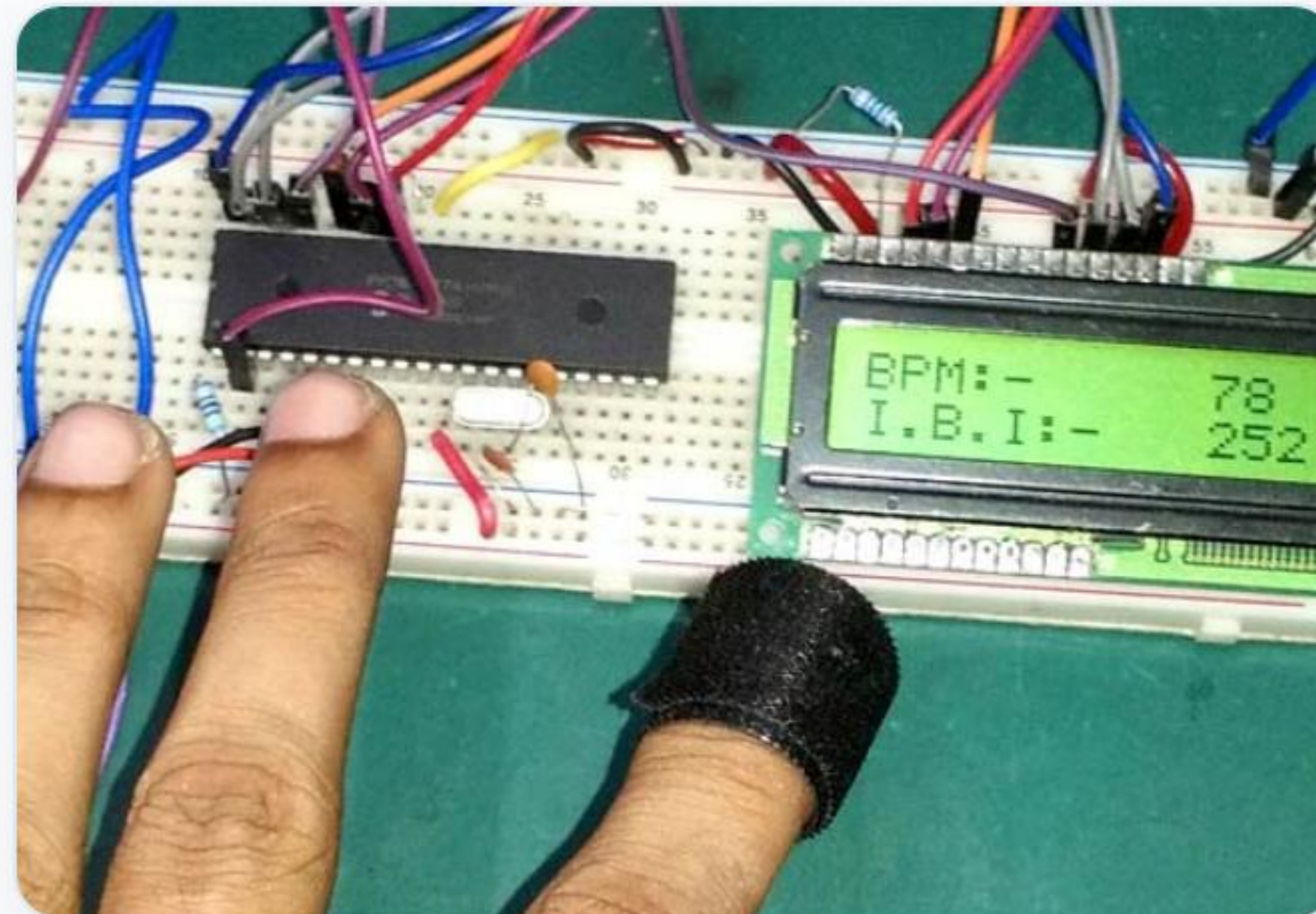
We have successfully established the engineering framework and core planning required for the EDI 0th Review:

- Formulated precise problem statement & requirement specifications.
- Finalized hardware architecture and component procurement.
- Completed initial 3D CAD chassis design and static load testing.
- Successfully breadboard-tested primary sensors and motor drivers.

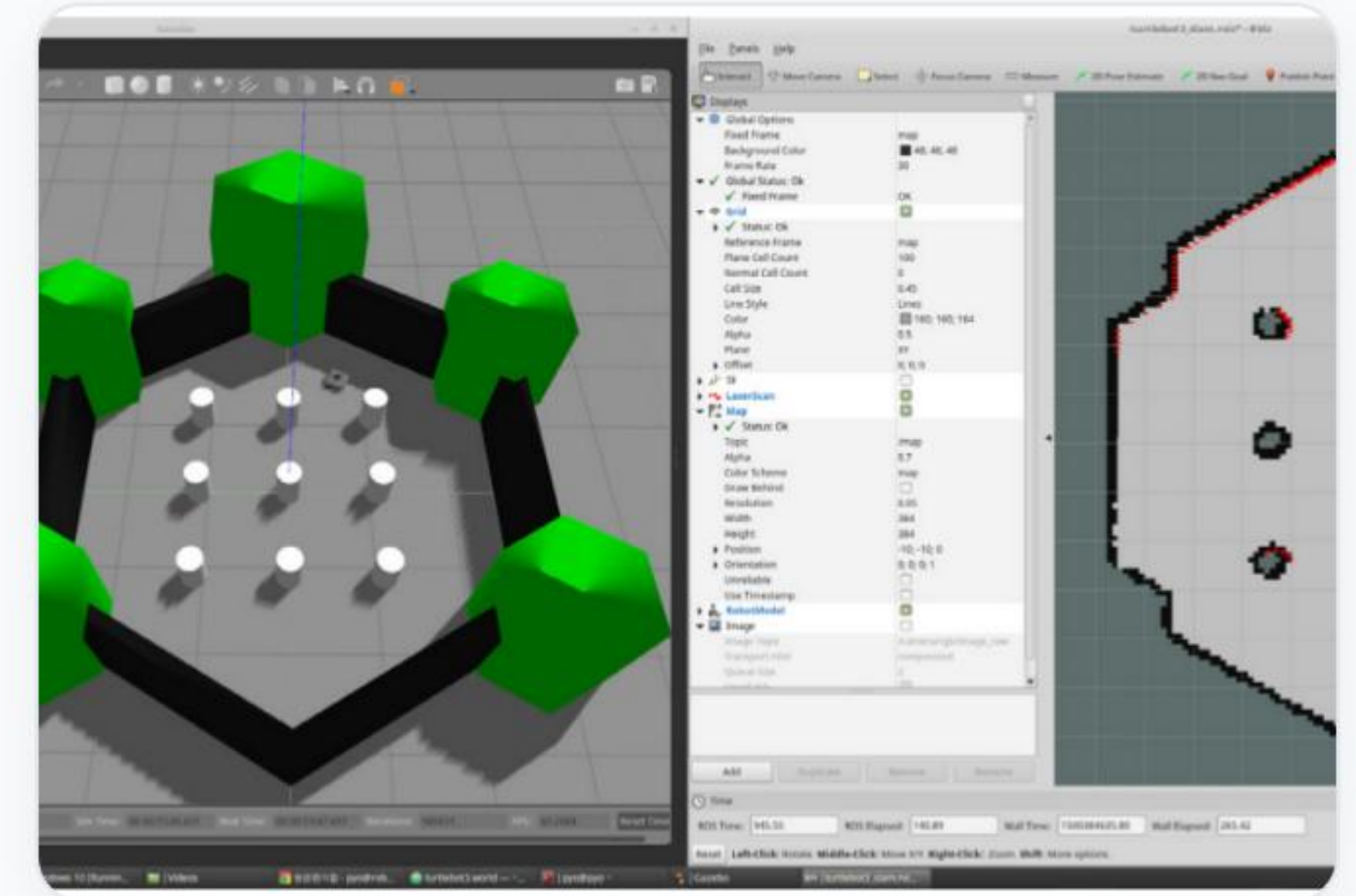
Prototyping & Simulation Progress

 Rescue robot 3D CAD model chassis design

Phase 1: 3D CAD Mechanical Chassis Design

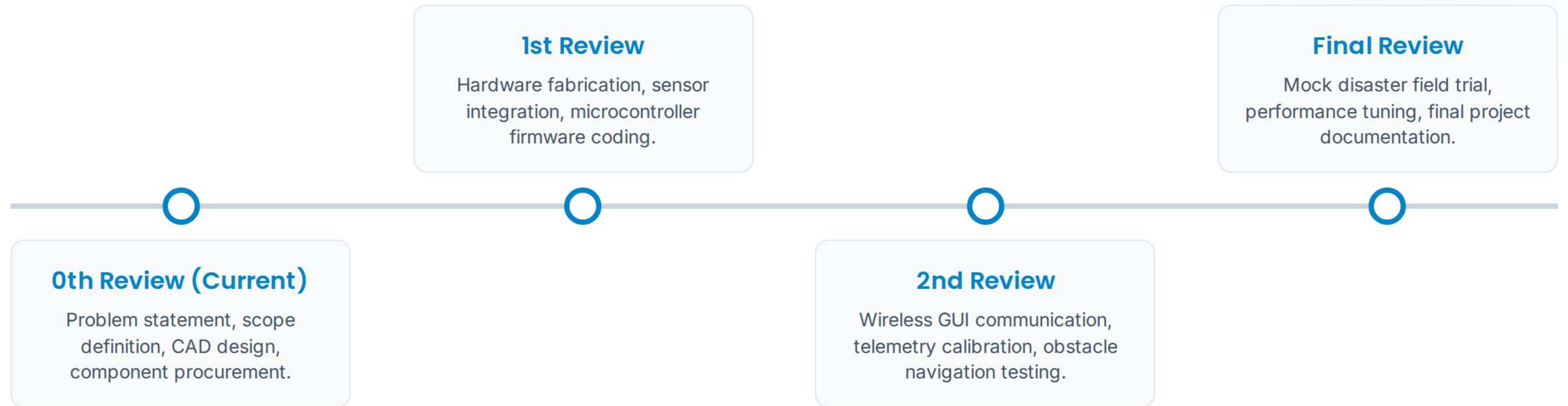


Phase 2: Microcontroller & Sensor Prototyping



Phase 3: Simulation Environment & Mapping Setup

Project Execution Roadmap



5. Core Conclusion & Philosophy

"Engineering innovation in robotics is not merely about mechanical automation, but extending human reach to preserve life when disaster strikes."

— Rescue Robotics EDI Project Team

Thank You!

Questions & Feedback for EDI 0th Review

Team Members: S Ritika | S Tanushri | S Vashnavi
Engineering Design & Innovation (EDI) Review

Image Sources



https://www.nist.gov/sites/default/files/styles/social/public/images/2017/02/07/rescue_robot_tests_to_offer_responders_high_tech_help.jpg?itok=tokGZ9Sc

Source: www.nist.gov



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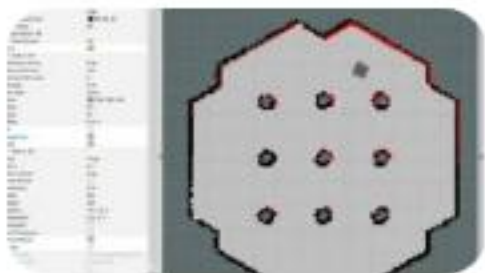
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